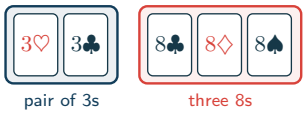
COMBINATORICS ON ONE PAGE

Partitioning a hand: Stirling, Bell and entropy

A five-card hand splits into groups of equal rank, and that pattern is a macrostate whose microstates are the actual hands.

1 / THE RULE AND THE PICTURE

A full house is a hand whose five cards split into a group of three and a group of two



pair of 3s — three 8s

The hand $3\heartsuit,\ 3\clubsuit,\ 8\clubsuit,\ 8\diamondsuit,\ 8\spadesuit$ sorts into the partition $\{3\heartsuit, 3\clubsuit \mid 8\clubsuit, 8\diamondsuit, 8\spadesuit\}$ of its five cards. The macrostate keeps only the pattern of shared ranks, here $(3, 2)$, and the microstates are all hands with that pattern:

$$S = k \ln W, \qquad W_{\text{full house}} = 3744$$

2 / THE STIRLING TRIANGLE / EACH ROW BUILT FROM THE ONE ABOVE

Row n counts the ways to group n cards by rank, and its sum is the Bell number B_n

$n=1$			1				$B_1 = 1$
$n=2$			1		1		$B_2 = 2$
$n=3$		1		3		1	$B_3 = 5$
$n=4$		1	7		6	1	$B_4 = 15$
$n=5$	1	15		25	10	1	$B_5 = 52$
$n=6$	1	31	90	65	15	1	$B_6 = 203$

$$S(n,k) = k\,S(n-1,k) + S(n-1,k-1)$$

Deal out one more card: it can join any of the k groups already formed, in k ways, or start a new group after $k-1$. So $S(5,3) = 3 \cdot 6 + 7 = 25$. The row $1, 15, 25, 10, 1$ is the five-card hand, with k the number of different ranks it shows, and the row sums $1, 2, 5, 15, 52, 203$ are the Bell numbers.

3 / PARTITIONING FIVE CARDS / THE STIRLING AND BELL COUNTS

The cards of a hand can be grouped by rank in $B_5 = 52$ ways, in seven patterns

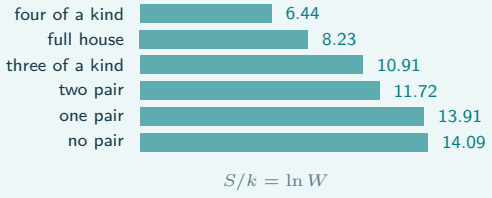
groups	pattern	name	partitions	how counted
x x x x x	(5)	five of a kind	1	all five alike, impossible in a real pack
x x x x y	$(4,1)$	four of a kind	5	choose the odd card out
x x x y y	$(3,2)$	full house	10	choose which two cards form the pair, $\binom{5}{2}$
x x x y z	$(3,1,1)$	three of a kind	10	choose the three alike, $\binom{5}{3}$
x x y y z	$(2,2,1)$	two pair	15	choose the single card, then split the other four into pairs in 3 ways
x x y z w	$(2,1,1,1)$	one pair	10	choose the pair, $\binom{5}{2}$
x y z w v	$(1,1,1,1,1)$	no pair	1	all five different

By number of groups, $S(5,2) = 5 + 10 = 15$ and $S(5,3) = 10 + 15 = 25$. The letters show only which cards share a rank.

4 / MICROSTATES AND ENTROPY / THE ACTUAL HANDS OF A REAL PACK

Choose the ranks, then the suits, and the count W of each pattern gives its entropy

pattern	name	rank choices	suit choices	W	P	S/k
$(4,1)$	four of a kind	$13 \cdot 12$	$1 \cdot 4$	624	0.024%	6.44
$(3,2)$	full house	$13 \cdot 12$	$4 \cdot 6$	3 744	0.144%	8.23
$(3,1,1)$	three of a kind	$13\binom{12}{2}$	$4 \cdot 4^2$	54 912	2.113%	10.91
$(2,2,1)$	two pair	$\binom{13}{2} \cdot 11$	$6^2 \cdot 4$	123 552	4.754%	11.72
$(2,1,1,1)$	one pair	$13\binom{12}{3}$	$6 \cdot 4^3$	1 098 240	42.257%	13.91
(1^5)	no pair	$\binom{13}{5}$	4^5	1 317 888	50.708%	14.09
total				$2\,598\,960 = \binom{52}{5}$	100%	



The full house takes the rank of the three in 13 ways and its suits in $\binom{4}{3} = 4$, then the pair's rank in 12 and suits in $\binom{4}{2} = 6$, so $13 \cdot 4 \cdot 12 \cdot 6 = 3744$. Five different ranks has the most microstates, and looking at suits and straights as well splits it into high card 1 302 540, straight 10 200, flush 5 108, straight flush 36 and royal flush 4, with $S/k = \ln 4 = 1.39$.

5 / WHERE THE STIRLING COUNT ENTERS / ORDERED DEALS

Partitions of the five positions times rank choices times suit orders gives every ordered deal

For a full house, 10 partitions of the positions times $13 \cdot 12 = 156$ rank choices times $4 \cdot 3 \cdot 2 \cdot 4 \cdot 3 = 288$ suit orders gives $449\,280 = 3744 \times 5!$, and summing over all seven patterns gives $311\,875\,200 = 52 \cdot 51 \cdot 50 \cdot 49 \cdot 48$, every ordered deal.

ASIDE / BELL AND THE PACK

The partitions of a five-card hand number $B_5 = 52$, the size of the pack, which looks like a coincidence.

LINKS TO OTHER SHEETS

Entropy, microstates and a growing pack of cards; *Cards, piles and finite differences*.

Convention. Hands are unordered sets of five cards from a pack of 52; ln is the natural logarithm and $k_B = 1$; (1^5) means five different ranks.

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